Classification of logic family

- Logic Families
  - can conduct electricity in 2 direction
    - Bipolar logic family (BJT)
      - Saturated
        - RTL (Resistor transistor logic)
        - DCTL (Direct coupled transistor logic)
        - DTL (Diode transistor logic)
        - TTL (transistor transistor logic)
      - Non saturated
        - Shorctly TTL
        - ECL
  - can conduct electricity in single direction
    - Unipolar logic family
      - PMOS
      - NMOS
      - CMOS

~~RTL → Resistor transistor logic~~

Characteristics of logic Families: / Important parameters of Digital IC :

① **Fan in :** The number of input of a ~~get~~ gate that can be handle without hamparing it's normal operation.

② **Fan out :** The number of gates that can be driven by gate output.

③ **Speed :** Determined by time between the application of input and change in output.

④ **Power dissipation :** Power used by per gate,

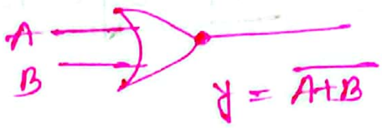
⑤ **Propagation Delay :** Time taken by the signal to reach to output from input.

⑥ **Noise margin :** Limit of Noise voltage ~~that can be present~~ that can be present without disturbing proper operation of circuit.

$\boxed{\text{RTL}}$ → Registor transistor logic.

RTL NOR GATE:

It work as ~~NAND~~ nor gate



$y = \overline{A+B}$

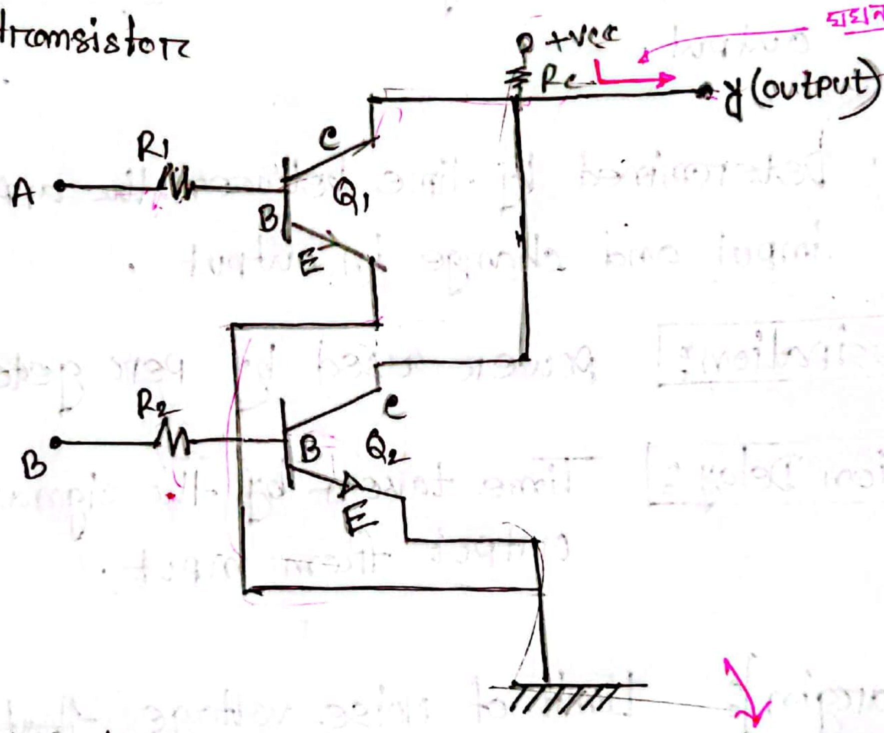
① with the help of resistor and transistor one logic gate is prepared which work as NOR gate.

A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

② Two resistor are connected with base on two transistor Q_1 & Q_2

③ Emitter of two transistor are grounded.

যখন ইনপুট 0 অর্থাৎ OPEN



~~RTL~~

Disadvantages:

(i) Switching speed is low due to registance.

(ii) Noise immunity is less.

(iii) power dissipation is more.

(iv) It is temperature sensitive.

ট্রান্সিস্টর 0 হলে OPEN
1 " Short

DTL → Diode transistor logic

With the help of diode and transistor one logic circuit is created which is called DTL. It behaves like (NAND/NOR) gate.

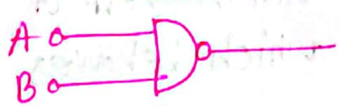
RTL-NAND GATE

Advantages:

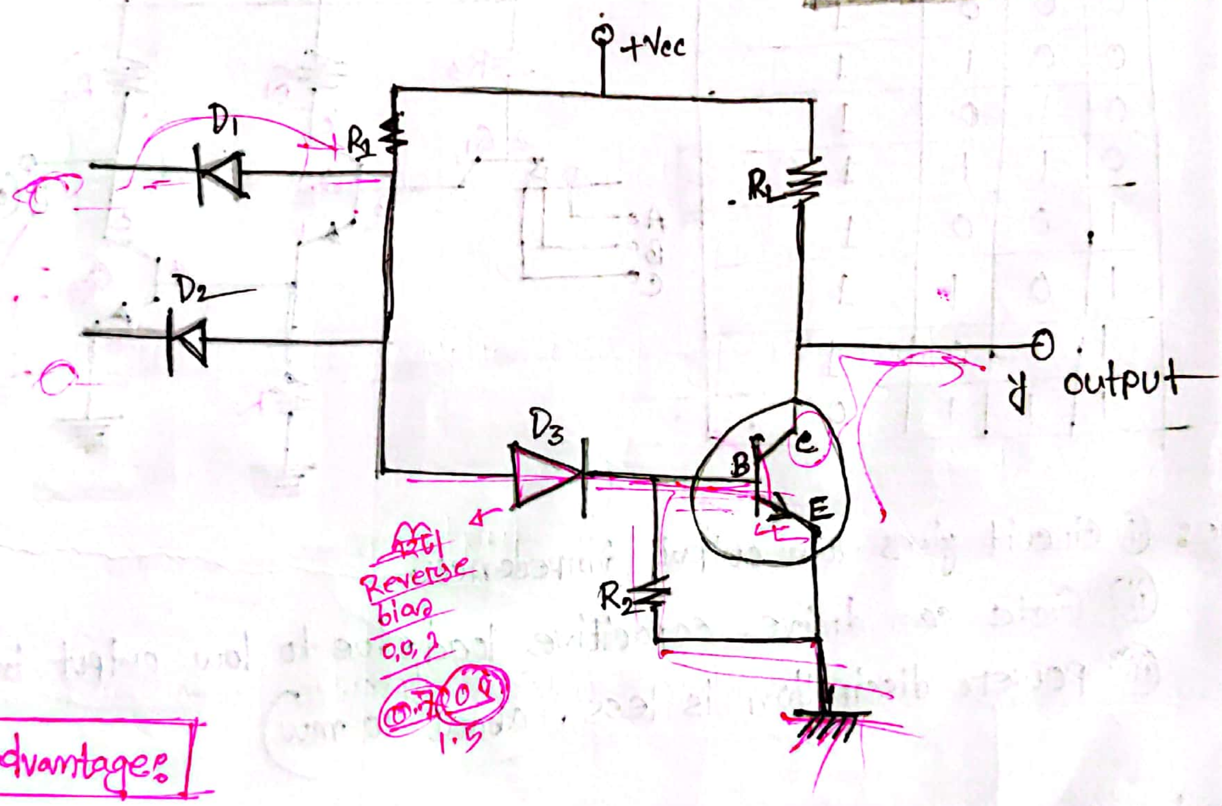
① output voltage is not reduced when load gate is increassed.

② Fan out is more.

③ Noise margin is more.



A	B	$\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0



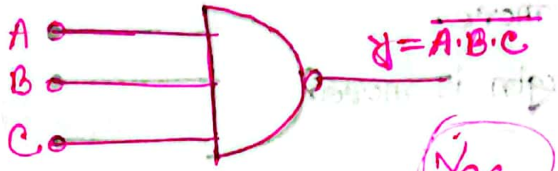
Disadvantage:

1. Propagation delay is more.
2. Due to this switching, speed is less.

TTL → Prop. delay $= \frac{1}{10 + 4}$ DTL

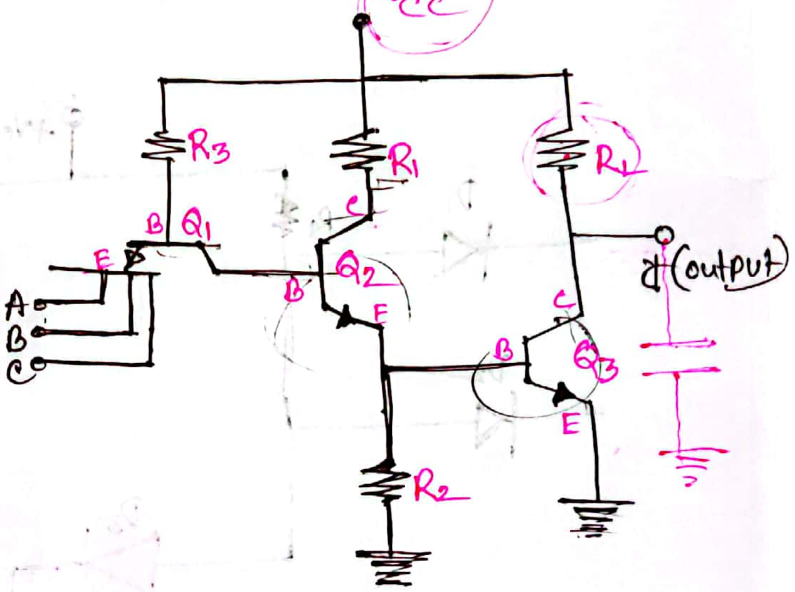
Transistor – transistor logic

→ With the help of transistor, A logic circuit is prepared which behaves like NAND gate.



multi emitter inverter

A	B	C	$y = \overline{A \cdot B \cdot C}$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



Advantage: (i) circuit gives low output impedence.

(ii) Gate can drive capacitive load due to low output impedence

(iii) Power dissipition is less. (about 10 mw)

Disadvantage:

(i) Large current spike is produced while switching.

CMOS → complement metal oxide semiconductor

* It behaves like inverter. (not)

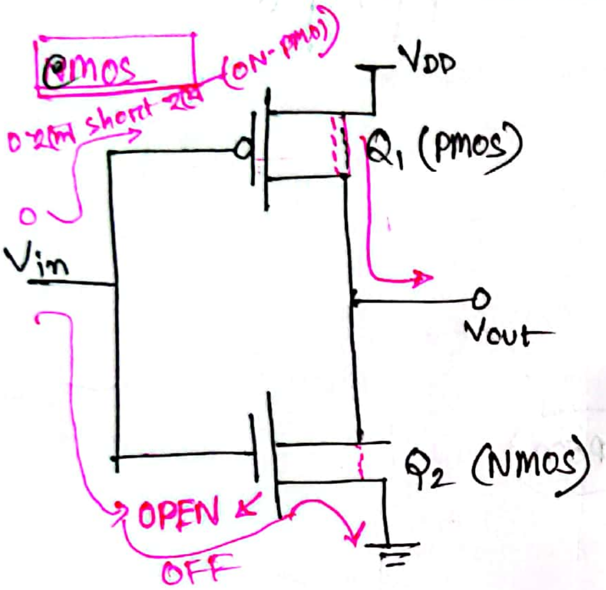
0 → 1
1 → 0

* CMOS = PMOS + NMOS

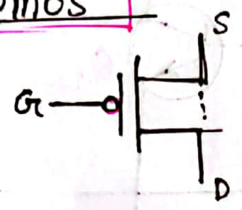
PMOS → P channel metal oxide semiconductor

NMOS → n channel metal oxide semiconductor

CMOS

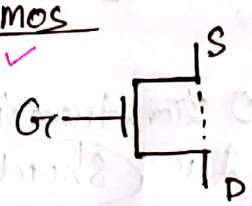


PMOS



G=0 → on → sc (short circuit)
G=1 → off → oc (open circuit)

NMOS



G = 0 → off → oc
G = 1 → on → sc

Vin	Q_1	Q_2	Vout
0	ON	OFF	1
1	OFF	ON.	0

Advantages:

1. stable for LSI (low scale integration)
2. Circuit is simple
3. Fabricated in small area.
4. power uses less.
5. work on 15V → 20V
6. Fan out is 50.
7. power dissipition 0.01 mW
8. Noise margin is 50V.

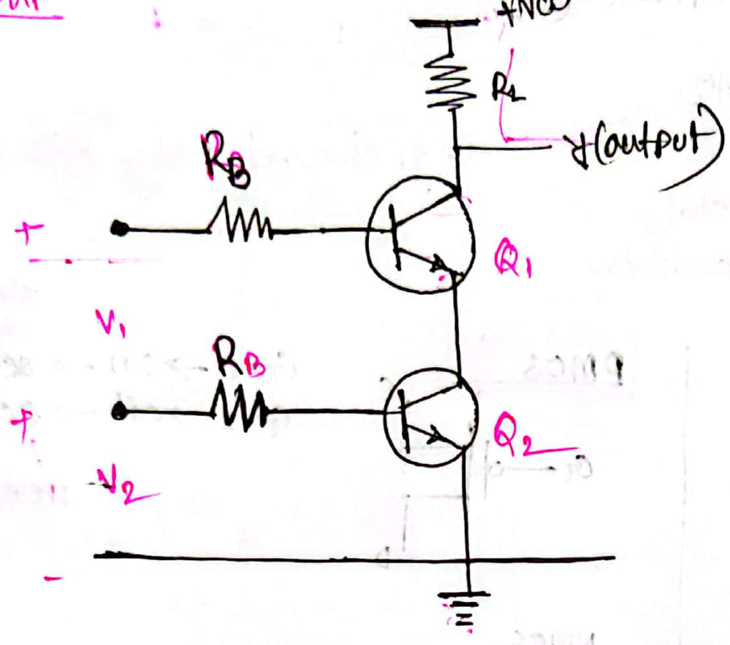
Disadvantages:

(i) Stored in special conducting form.
(ii) Propagation time is more (70 ms)
(iii) Cost is more than TTL.

$V_1 \cdot V_2 \Rightarrow \overline{V_1 \cdot V_2}$

RTL NAND GATE

2 input

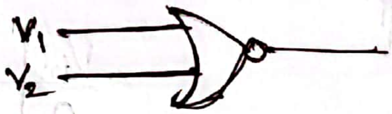


V_1	V_2	$y = \overline{V_1 \cdot V_2}$
0	0	1
0	1	1
1	0	1
1	1	0

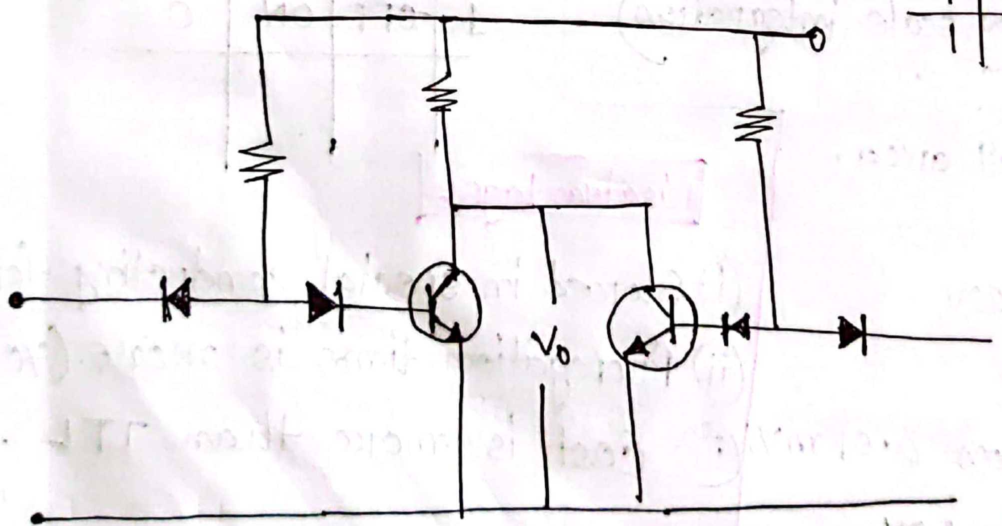
0 হলে, transistor open
1 " short

2 input DTL-NOR Gate

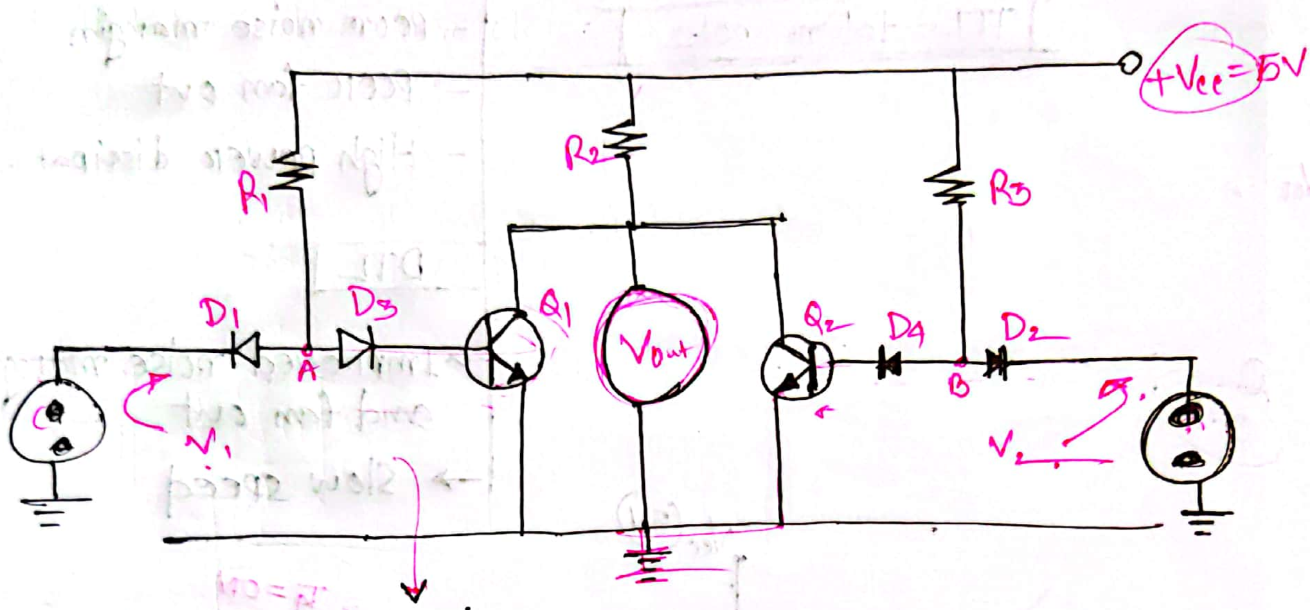
$\boxed{\text{Diode - transistor logic}}$



V_1	V_2	$\overline{V_1 + V_2}$
0	0	1
0	1	0
1	0	0
1	1	0



Again draw:



$V_A = \cancel{0.7+}\ V_{D_3} + V_{BE_1}$

$= 0.7 + 0.8$

$\Rightarrow 1.5V$ হলে, তবেই transistor on অবস্থায় থাকবে

$V_B = V_{D_4} + V_{BE_2}$

$= 0.7 + 0.8$

$= 1.5V$

$V_1 = 0, V_2 = 0$ হলে, D_1 ও D_2 এর মধ্যে দিয়ে votage যাবে

Q_1 and Q_2 will be open

$\therefore V_{out} = 1$

$V_1 = 1, V_2 = 0$, D_1 will be open. and Q_1 will be short.

$\therefore V_{out} = 0$

Simillers.

pros & cons

~~Operation Amplifier~~

TTL- totem pole

RTL

- poor noise margin
- poor fan out
- High power dissipat.

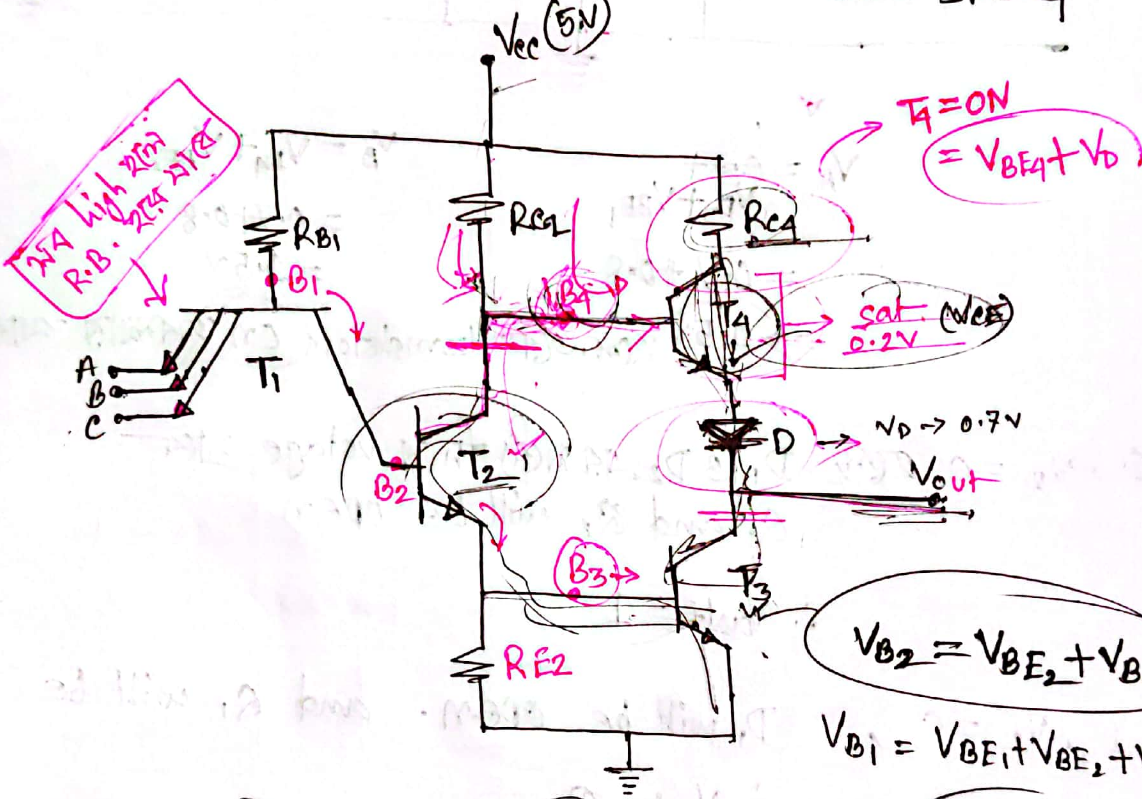
DTL

→ Improved noise margin and fan out

→ Slow speed

V_{CE} →

0.2



$$V_{B_2} = V_{BE_2} + V_{BE_3}$$

$$V_{B_1} = V_{BE_1} + V_{BE_2} + V_{BE_3}$$

এখানে, যদি A, B ও C তে লো সিগন্যাল দেওয়া হয় তাহলে T_2 ও T_3 অফ অবস্থায় (OPEN) থাকবে এক্ষেত্রে T_4 অন অবস্থায় থাকবে,

$T_2, T_3 = OFF$

$T_4 = ON$

$$\therefore V_0 = V_{CC} - R_{C4}(I_{C4}) - \underset{(0.2)}{V_{CE4}} - \underset{(0.7)}{V_D}$$

Operational Amplifier

এই ক্ষেত্রে, সব input on (high) হলে, T_2 ও T_3 on state এ থাকবে,

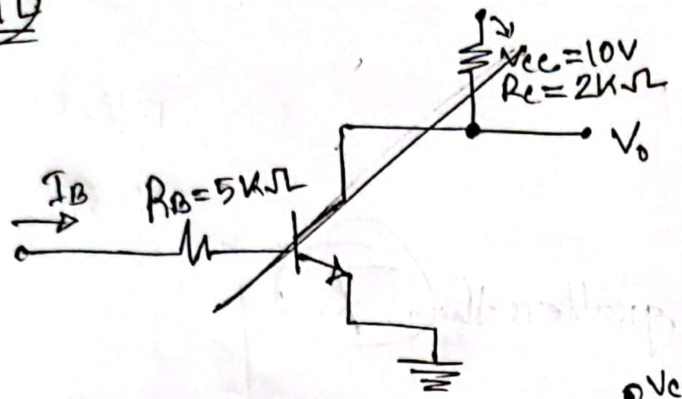
এক্ষেত্রে $$V_{B4} = V_{CE_2} + V_{B4}$$

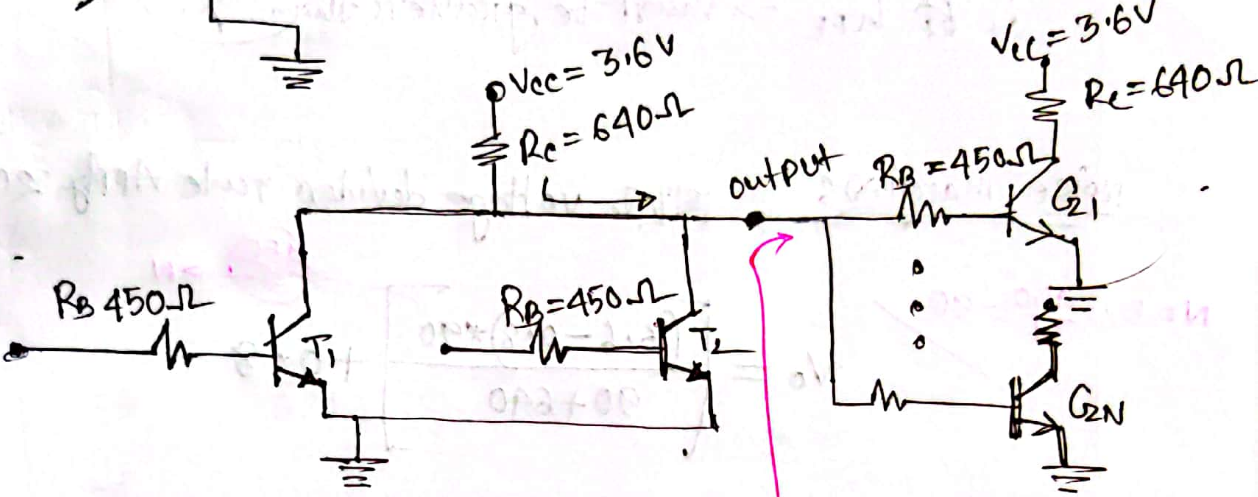
D, ডায়োড টি ব্যবহৃত হয় কারন, যখন $T_3 = ON$ তখন $T_4 = OFF$

আবার $T_3 = OFF$ " $T_4 = ON$

RTL

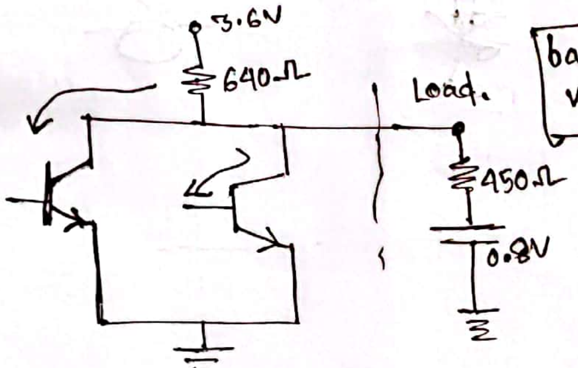
3.2





When, input is low → output is high → তখন এইটা আবার reverse.

then we can draw it as.



base-emitter voltage drop = 0.8V

সমান্তরালে থাকায় voltage drop same.

we know,

$$\boxed{h_{FE}\cdot I_B \geq I_c}$$

$V=IR \Rightarrow I=\frac{V}{R}$

Base current for each ~~every~~ load transistor:

$$\therefore I_B = \left(\frac{3.6-0.8}{640+\frac{450}{N}}\right)\frac{1}{N}$$

$$= \frac{2.8}{640N+450}, \quad N=5, \quad \therefore I_B = 0.767\text{ mA}.$$

The collector current for the load transistor in saturation

$$I_{c\,sat} = \frac{3.6-0.2}{640} = 5.31\text{ mA}$$

let, $h_{FE} \cdot I_B = I_C$

$\Rightarrow h_{FE} * 0.767 \geq 5.31$

$\Rightarrow h_{FE} \geq 7$

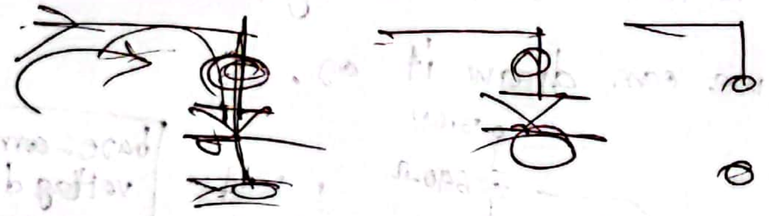
$\therefore$ h_{FE} → must be greater than 7.

Noise margins: এখানে voltage devider rule Apply হবে

$N=5, \frac{450}{N} = 90$

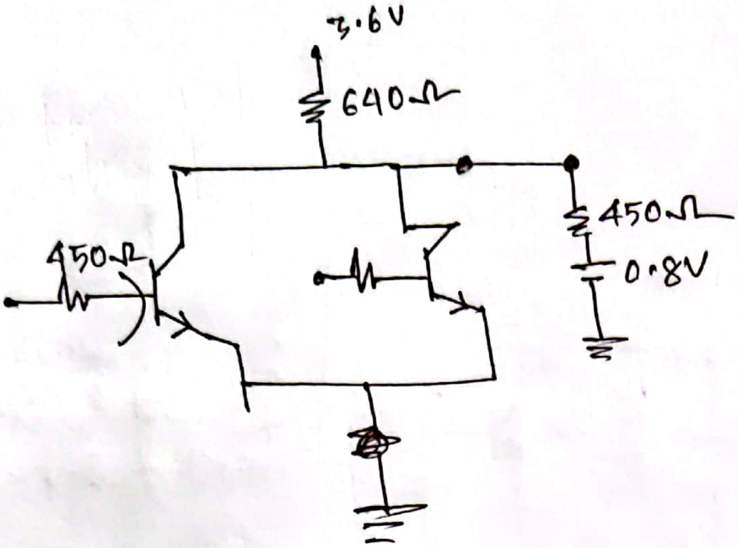
$\frac{450}{5} \rightarrow N$

$$V_o = \left[\frac{(3.6-0.8)*90}{90+640}\right] + 0.8$$



Practice

RTL



$$I_B = \frac{(3.6-0.8)}{640+\frac{450}{N}} * \frac{1}{N}$$

$$= \frac{2.8}{640N+450}$$

Given,

$N = 5$

$\therefore \boxed{I_B = 0.767\ mA}$

$h_{FE} * I_B \geq I_C$

$\therefore h_{FE} \geq \frac{5.31}{0.767}$

$\boxed{h_{FE} \geq 7}$

$$I_{C_{SAT}} = \frac{3.6-0.2}{640}$$

$$= 5.31\ mA$$

Noise margins

$$V_o = \frac{(3.6-0.8)*90}{90+640} + 0.8$$

$$= 1.145$$

??

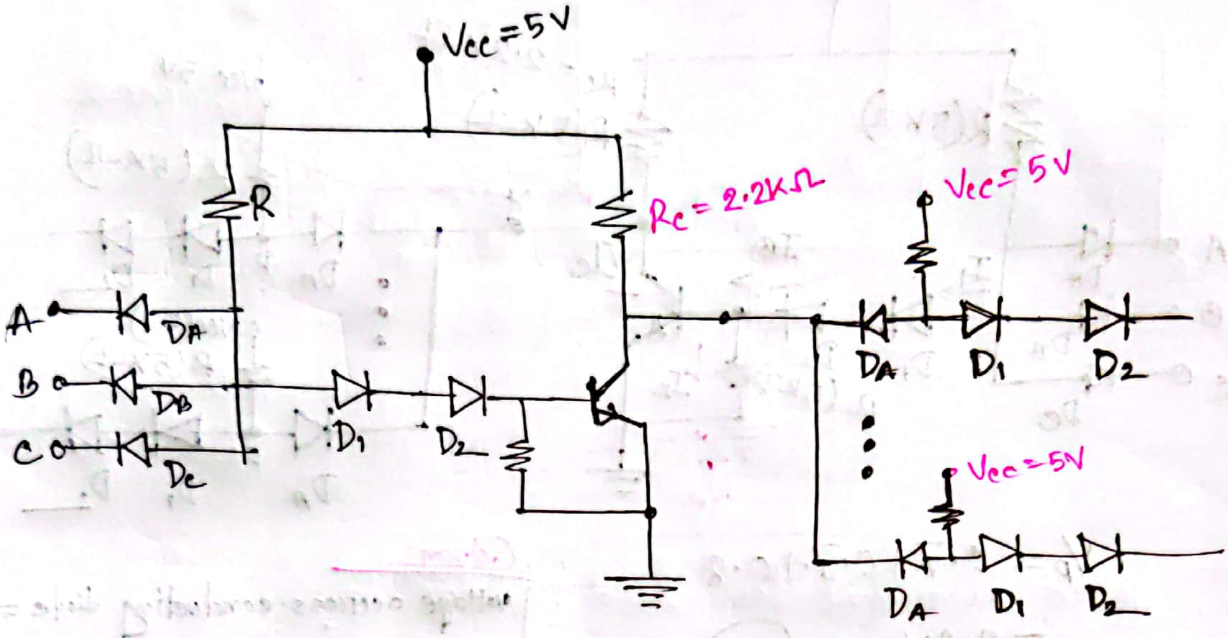
For,

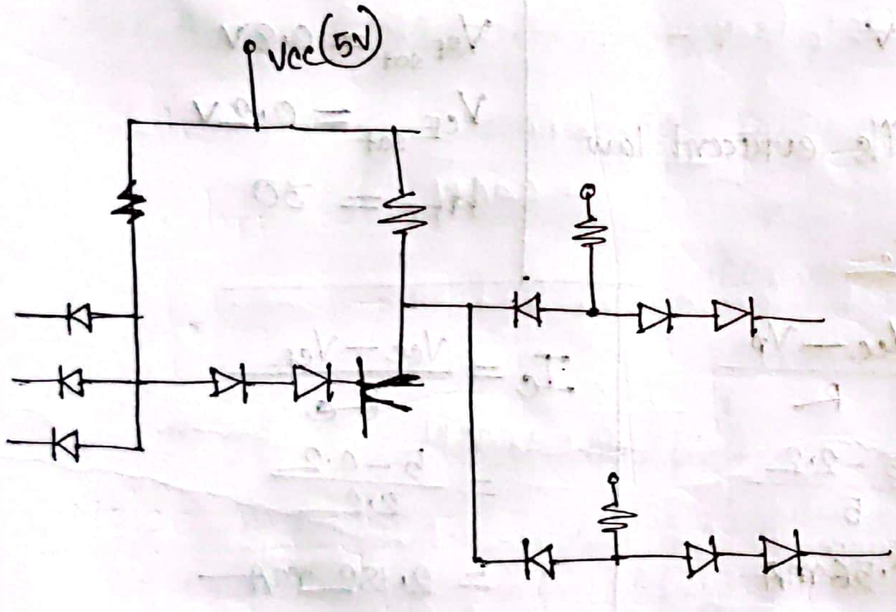
$h_{FE} = 10$

$h_{FE} * I_B \geq I_C$

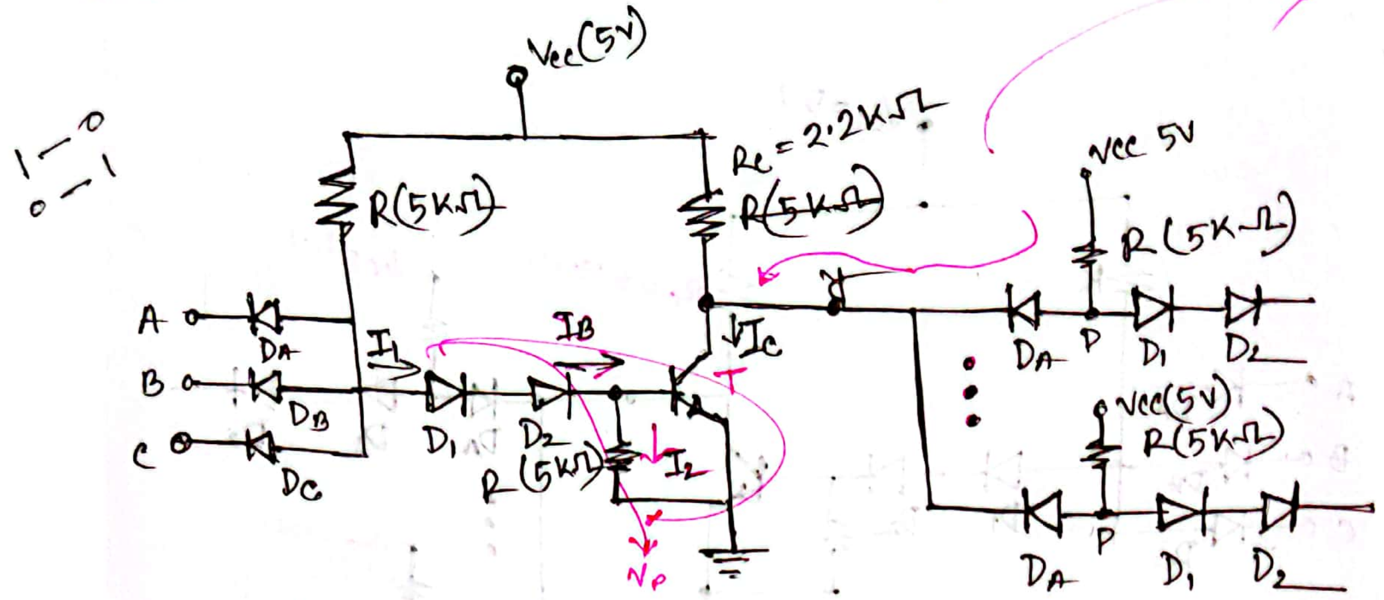
$$I_B = \frac{I_C}{h_{FE}} * 5 = \frac{5.31}{10} * 5$$

DIODE Transistor Logic





DTL



$V_p = 0.7 + 0.7 + 0.8$

$= 2.2V$

If all input are high:

Low level (0) $= 0.2V$

High " $V(1) = 5V$

Given

voltage across conducting diode = 0.

cut in voltage = 0.6V

Transistor: Cut in voltage = 0.5

$V_{BE\,sat} = 0.8V$

$V_{CE\,sat} = 0.2V$

$h_{FE} = 30$

writing kirchhoff's current law

$I_B = I_1 - I_2$

where, $I_1 = \frac{V_{cc} - V_p}{R}$

$= \frac{5 - 2.2}{5}$

$= 0.56mA$

$I_2 = \frac{0.8}{5}$

$= 0.16mA$

$\therefore I_B = (0.56 - 0.16)\,mA$

$= 0.4\,mA$

$I_c = \frac{V_{cc} - V_{CE}}{R_c}$

$= \frac{5 - 0.2}{2.2}$

$= 2.182\,mA$

$h_{FE} * I_B > I_c$

$\Rightarrow \cancel{12}\ 30 \times .4 > 2.182$

$\Rightarrow 12 > 2.182$

$\therefore$ It is confirmed that transistor are in saturation and.

Now,
$$I_L = \frac{5-(0.7+0.2)}{5}$$
$$= 0.82mA$$

N Load gates are fed from the gate :

~~$I_L \times N + I_C \leq$~~ $I_C \leq h_{FE} * I_B$

$\Rightarrow$ $N * I_L + 2.182 = 12$

$\Rightarrow N = \frac{12 - 2.182}{0.82}$

$\therefore N \leq 12$, $\boxed{N = 10}$

(ii) If at least one of the input is low,

the $V_P = 0.2 + 0.7 = 0.9V$

minimum voltage required for D_1, D_2 and $F = 0.6 + 0.6 + 0.5$
$= 1.7V$

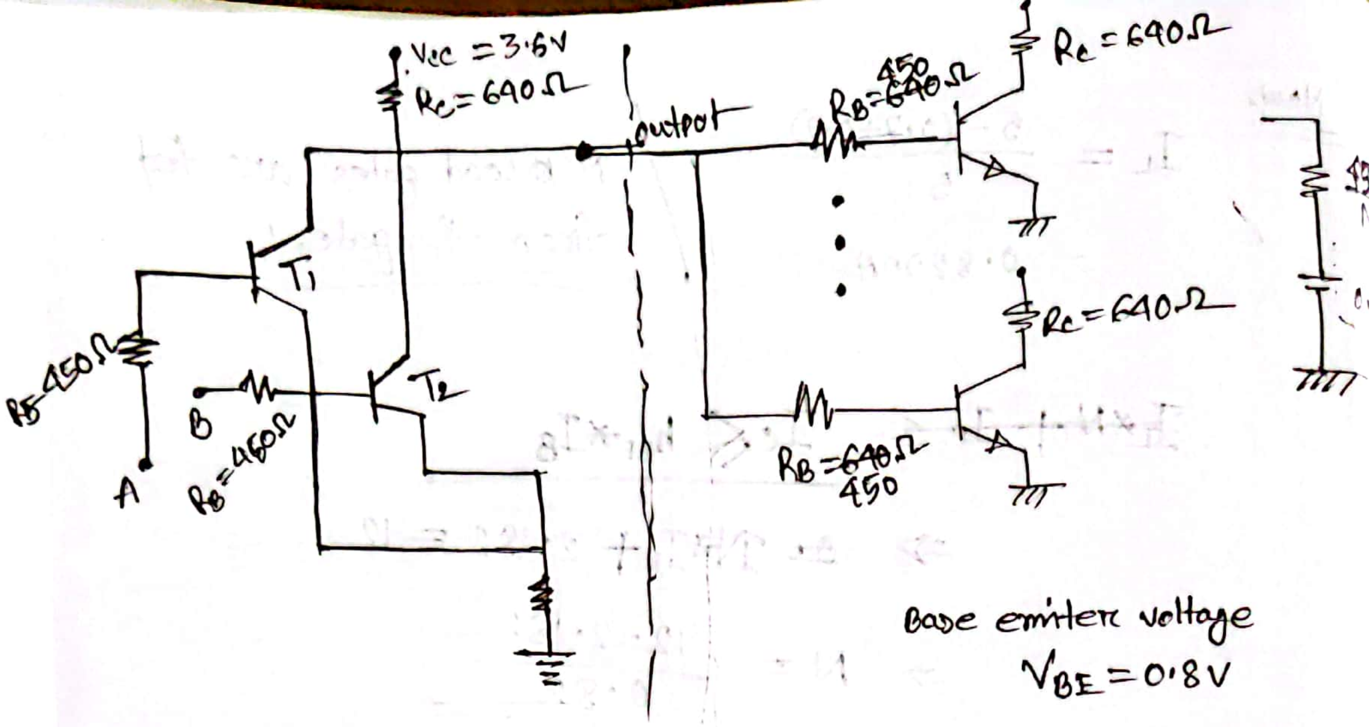
non conducting
output is $\Rightarrow$ 5V

⊛ If all the inputs are high, the output is low. Since $V_P = 2.2V$, the input diode are in reverse ~~diode~~ biased $5 - 2.2 = 2.8$.

cut voltage of the diode is $\Rightarrow 0.6V$

$\therefore$ A negative ~~voltage~~ noise spike at least

3.4V



Base emitter voltage
$V_{BE} = 0.8V$

∴ Base current for load resistor is

$$\left(640 \times I_B\right) + \left(\frac{450}{N} \times I_B\right) + 0.8V = 3.6$$

$$\Rightarrow \cancel{640} I_B \left(640 + \frac{450}{N}\right) = 2.8$$

$$\Rightarrow I_{B\,(total)} = \frac{2.8N}{450 + 640N}$$

The base current for every load transistor.

$$I_B = \frac{2.8}{450 + 640N}$$

The collector current for the load transistor

$$640 I_C + 0.2 = 3.6$$

$$I_{C_{SAT}} = \frac{3.4}{640} = 5.31 \times 10^{-3}$$

collector emitter voltage = 0.2

The value of N must be satisfy following relation.

$$I_B \times h_{FE} \geqslant I_{C_{sat}}$$

$$I_B = \frac{2.8}{450 + 640(5)}$$
$$= 0.767\ mA$$

$$\therefore h_{FE} = \frac{5.31}{0.767}$$
$$= 6.92$$

$\therefore$ Hence, h_{FE} must be greater than 7

$\frac{5}{640}$

$\frac{5}{450}+$

$N = 5$

Noise margin:

Logic: 0 When the output is in 0 state $V_0 = 0.2V$

If this voltage become about 0.5V (cut in voltage of transistor) the load transistor comes to conduction.

Here the logic 0, noise margin $= (0.5 - 0.2) = 0.3V$

Logic: 1 It depend upon the number of gates.

$$V_0 = \frac{90}{640+90}(3.6) + \frac{640}{90+640}(0.8)$$
$$= 1.14$$

For $h_{FE} = 10$, the total current required for load transistor

$$I_B = N\frac{I_{C_{SAT}}}{h_{FE}} = 5 \times \frac{5.31}{10} = 2.655\ mA$$

$V = IR \Rightarrow$ Total voltage drop $= (2.655 \times 10^{-3} \times 90) = 1.04V$

$\therefore$ Noise margin for level 1.

$\Delta_1 = (1.14 - 1.04)$
$= 0.1V$

Propagation delay $= \left(640 + \frac{450}{N}\right)$